Mathematical Statistics I
HOMEWORK 8

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1.

$$Y \sim Beta(\alpha = 1, \beta = 3), \qquad 0 \le Y \le 1$$

$$f_Y(y) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} y^{\alpha-1}(1-y)^{\beta-1} = \frac{\Gamma(4)}{\Gamma(1)\Gamma(3)} y^0(1-y)^2 = \frac{3!}{0!\,2!} y^0(1-y)^2$$

$$f_Y(y) = \frac{6}{1 \cdot 2} y^0(1-y)^2 = 3(1-y)^2$$

$$F_Y(y) = \int_0^y 3(1-y)^2 dy = [-(1-y)^3]_0^y = 1-(1-y)^3$$

$$P(Y \le y_0) = F_Y(y_0) = 0.90$$

$$1-(1-y_0)^3 = 0.90$$

$$(1-y_0)^3 = 0.10$$

$$1 - y_0 = 0.10^{\frac{1}{3}}$$

$$y_0 = 1 - 0.10^{\frac{1}{3}}$$

$$y_0 = 0.535841117$$

$$Budget = 1000y_0 \sim 1000 \cdot 0.535841117 \sim \$535.84$$

The money that should be budgeted for repair costs so the actual cost will exceed the budget amount only 10% of the time is about \$536.

2.

a.

$$Y \sim Beta(\alpha = 2, \beta = 2), \qquad 0 \le Y \le 1$$

$$f_Y(y) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} y^{\alpha-1}(1-y)^{\beta-1} = \frac{\Gamma(4)}{\Gamma(2)\Gamma(2)} y^1(1-y)^1 = \frac{3!}{1!\,1!} y(1-y)$$

$$f_Y(y) = \frac{6}{1 \cdot 1} y(1-y) = 6y(1-y)$$

$$F_Y(y) = \int_0^y 6(1-y)^2 dy = 6\left[\frac{y^2}{2} - \frac{y^3}{3}\right]_0^y = 6\left(\frac{y^2}{2} - \frac{y^3}{3}\right) - 6\left(\frac{0^2}{2} - \frac{0^3}{3}\right) = 3y^2 - 2y^3$$

$$F_Y(y) = 3y^2 - 2y^3$$

b.

$$F_Y(y) = 3y^2 - 2y^3$$

$$\mathrm{P}(0.5 \le \mathrm{Y} \le 0.8) = F_Y(0.8) - F_Y(0.5)$$

$$F_Y(0.8) = 3(0.8)^2 - 2(0.8)^3$$

$$F_Y(0.5) = 3(0.5)^2 - 2(0.5)^3$$

$$F_Y(0.8) = 3(0.64) - 2(0.512) = 1.92 - 1.024 = 0.896$$

$$F_Y(0.5) = 3(0.25) - 2(0.125) = 0.75 - 0.25 = 0.50$$

$$\mathrm{P}(0.5 \le \mathrm{Y} \le 0.8) = 0.896 - 0.50 = 0.396$$

$$\mathrm{P}(0.5 \le \mathrm{Y} \le 0.8) = 0.396$$

3.

$$\binom{9}{3} = 84$$

$$4 \text{ married } \binom{4}{y_1}$$

$$3 \text{ never-married } \binom{3}{y_2}$$

$$3 - y_1 - y_2 \text{ from 2 divorced: } \binom{2}{3 - y_1 - y_2}$$

$$P(Y_1 = y_1, Y_2 = y_2) = \frac{\binom{4}{y_1}\binom{3}{y_2}\binom{2}{3 - y_1 - y_2}}{\binom{9}{3}}$$

$$y_1 \in \{0,1,2,3\}\ and\ y_1 \le 4$$

$$y_2 \in \{0,1,2,3\}\ and\ y_2 \le 3$$

$$3 - y_1 - y_2 \in \{0,1,2\}$$

$$y_1 + y_2 \leq 3$$

$$\binom{9}{3} = 84$$

$Y_1 = 0$

$$(0,0): \binom{2}{3} = 0$$

$$(0,0): \frac{\binom{4}{0}\binom{3}{0}\binom{2}{3}}{84} = 0$$

$$(0,1): \frac{\binom{4}{0}\binom{3}{1}\binom{3}{2}}{84} = \frac{1 \cdot 3 \cdot 1}{84} = \frac{3}{84} = 0.035714$$

$$(0,2): \frac{\binom{4}{0}\binom{3}{2}\binom{2}{1}}{84} = \frac{1 \cdot 3 \cdot 2}{84} = \frac{6}{84} = 0.071429$$

$$(0,3): \frac{\binom{4}{0}\binom{3}{3}\binom{2}{0}}{84} = \frac{1 \cdot 1 \cdot 1}{84} = \frac{1}{84} = 0.011905$$

$Y_1 = 1$

$$(1,0): \frac{\binom{4}{1}\binom{3}{0}\binom{2}{2}}{84} = \frac{4 \cdot 1 \cdot 1}{84} = \frac{4}{84} = 0.047619$$

$$(1,1): \frac{\binom{4}{1}\binom{3}{1}\binom{2}{1}}{84} = \frac{4 \cdot 3 \cdot 2}{84} = \frac{24}{84} = 0.285714$$

$$(1,2): \frac{\binom{4}{1}\binom{3}{2}\binom{2}{0}}{84} = \frac{4 \cdot 3 \cdot 1}{84} = \frac{12}{84} = 0.142857$$

$$(1,3): 0$$

$Y_1 = 2$

$$(2,0): \frac{\binom{4}{2}\binom{3}{0}\binom{2}{1}}{84} = \frac{6 \cdot 1 \cdot 2}{84} = \frac{12}{84} = 0.142857$$

$$(2,1): \frac{\binom{4}{2}\binom{3}{1}\binom{2}{0}}{84} = \frac{6 \cdot 3 \cdot 1}{84} = \frac{18}{84} = 0.214286$$

$$(2,2): 0$$

$$(2,3): 0$$

$Y_1 = 3$

$$(3,0): \frac{\binom{4}{3}\binom{3}{0}\binom{2}{0}}{84} = \frac{4 \cdot 1 \cdot 1}{84} = \frac{4}{84} = 0.047619$$

$$(3,1): 0$$

$$(3,2): 0$$

$$(3,3): 0$$

Y_1/Y_2	0	1	2	3
0	0	0.035714	0.071429	0.011905
1	0.047619	0.285714	0.142857	0
2	0.142857	0.214286	0	0
3	0.047619	0	0	0

4.

a.

$$(0.38, 0.17, 0.14, 0.02, 0.24, 0.05) \geq 0$$

$$0.38 + 0.17 + 0.14 + 0.02 + 0.24 + 0.05 = 1$$

Verified that this is a legitimate probability distribution due to values are non-negative and table sums to 1.

b.

$$F(1,1) = P(Y_1 \leq 1, Y_2 \leq 1) = \sum_{y_1=0}^{1} \sum_{y_2=0}^{1} p(y_1, y_2)$$

Y_2/Y_1	0	1
0	0.38	0.17
1	0.14	0.02

$$F(1,1) = 0.38 + 0.17 + 0.14 + 0.02 = 0.71$$

$$F(1,1) = 0.71$$

c.

Y_2/Y_1	0	1
0	0.38	0.17
1	0.14	0.02
2	0.24	0.05

$$Y_1 = 0$$

$$p_{Y1}(0) = 0.38 + 0.14 + 0.24 = 0.76$$

$$p_{Y1}(1) = 0.17 + 0.02 + 0.05 = 0.24$$

$$p_{Y2}(0) = 0.38 + 0.17 = 0.55$$

$$p_{Y2}(1) = 0.14 + 0.02 = 0.16$$

$$p_{Y2}(2) = 0.24 + 0.05 = 0.29$$

Y_1:

$$p_{Y1}(0) = 0.76$$

$$p_{Y1}(1) = 0.24$$

Y_2:

$$p_{Y2}(0) = 0.55$$

$$p_{Y2}(1) = 0.16$$

$$p_{Y2}(2) = 0.29$$

5.

a.

$$f(x,y) = x + y, 0 \le x \le 1, y \le 1$$

$$0 \le x < \frac{1}{2}$$

$$\frac{1}{4} < y \le 1$$

$$P\left(X < \frac{1}{2}, Y > \frac{1}{4}\right) = \int_{x=0}^{1/2} \int_{y=1/4}^{1} (x+y)dydx$$

$$\int_{1/4}^{1} (x+y)dy = \left[xy + \frac{y^2}{2}\right]_{y=1/4}^{1} = \frac{3}{4}x + \frac{15}{32}$$

$$\int_{1/4}^{1/2} \left(\frac{3}{4}x + \frac{15}{32}\right) dx = \frac{3}{4}\left[\frac{x^2}{2}\right]_0^{1/2} + \frac{15}{32}[x]_0^{1/2} = \frac{3}{32} + \frac{15}{64} = \frac{21}{64}$$

$$P\left(X < \frac{1}{2}, Y > \frac{1}{4}\right) = \frac{21}{64} \sim 0.3281$$

b.

$$f(x,y) = x + y, 0 \le x \le 1, y \le 1$$

$$P(X + Y < 1) = \int_{x=0}^{1} \int_{y=0}^{1-x} (x+y)dydx$$

$$\int_{0}^{1-x} (x+y)dy = x(1-x) + \frac{(1-x)^2}{2} = \frac{1-x^2}{2}$$

$$\int_{0}^{1} \frac{1-x^2}{2} dx = \frac{1}{2}\left[x - \frac{x^3}{3}\right]_0^1 = \frac{1}{2}\left(1 - \frac{1}{3}\right) = \frac{1}{3}$$

$$P(X + Y < 1) = \frac{1}{3} \sim 0.3333$$

6.

a.

$$f(x,y) = e^{-(x+y)}, \quad x, y > 0$$

$$P(X < 1, Y > 5) = \int_0^1 \int_5^{\infty} e^{-(x+y)} dydx = \int_0^1 e^{-x}[e^{-y}]_{y=5}^{\infty} dx = \int_0^1 e^{-x}e^{-5} dx = e^{-5}[-e^{-x}]_0^1$$

$$P(X < 1, Y > 5) = e^{-5}(1 - e^{-1}) \sim 0.00426$$

$$P(X < 1, Y > 5) \sim 0.00426$$

b.

$$f(x, y) = e^{-(x+y)}, \quad x, y > 0$$

$$P(X + Y < 3) = \int_0^3 \int_5^{3-x} e^{-(x+y)} dy dx = \int_0^3 e^{-x}[-e^{-y}]_0^{3-x} dx = \int_0^3 e^{-x}(1 - e^{-(3-x)}) dx$$

$$P(X + Y < 3) = \int_0^3 e^{-x} dx - e^{-3} \int_0^3 1 dx = (1 - e^{-3}) - 3e^{-3} = 1 - 4e^{-3} \sim 0.80085$$

$$P(X + Y < 3) \sim 0.80085$$

c.

$$f(x, y) = e^{-(x+y)}, \quad x, y > 0$$

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

$$f_X(x) = \int_0^{\infty} e^{-(x+y)} dy = e^{-x} \int_0^{\infty} e^{-y} dy = e^{-x}[-e^{-y}]_0^{\infty} = e^{-x}(1 - 0) = e^{-x}$$

$$f_Y(y) = \int_0^{\infty} e^{-(x+y)} dx = e^{-y} \int_0^{\infty} e^{-x} dx = e^{-y}(1 - 0) = e^{-y}$$

$$f_X(x) = e^{-x}, f_Y(y) = e^{-y}$$

$$X \sim Exp(1) \quad and \quad Y \sim Exp(1)$$

d.

$$f(x, y) = e^{-(x+y)}, \quad x, y > 0$$

$$P(1 < X < 2.5) = F(2.5) - F(1) = (1 - e^{-2.5}) - (1 - e^{-1}) = e^{-1} - e^{-2.5}$$

$$P(1 < X < 2.5) = 0.367879 - 0.082085 \sim 0.285794$$

$$P(1 < X < 2.5) \sim 0.28579$$

Y has the same distribution

$$P(1 < Y < 2.5) = e^{-1} - e^{-2.5} \sim 0.28579$$

$$P(1 < X < 2.5) = P(1 < Y < 2.5) = e^{-1} - e^{-2.5} \sim 0.28579$$